

MOHAMED MAGDY ELATTAR

AI Engineer | Software Engineer | Machine Learning & Data Science

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PROFILE

Computer Science graduate and independent builder specializing in applied machine learning, agentic AI, and full-stack SaaS product development. Sole technical lead and architect across a broad portfolio spanning a multi-tenant pharmacy SaaS platform, multi-agent AI systems, IoT platforms, and computer-vision models. Strong foundation in Python, SQL, TensorFlow, LLM API integration, and cloud/IoT deployment, with a self-driven approach to designing, architecting, and building working products end to end.

INDEPENDENT PROJECTS & FREELANCE WORK

Founder / Lead Developer — Self-Directed

2023 – Present

Meridian — Multi-Branch Pharmacy SaaS Platform — React, Supabase (Postgres), Vercel, Row-Level Security

- Architected and built a multi-tenant SaaS platform for Egyptian multi-branch pharmacy chains, migrating the system from an initial Firebase/Firestore prototype to a relational Postgres/Supabase stack to support complex role-based access and reporting.
- Designed a seven-tier role hierarchy enforced end-to-end with Postgres Row-Level Security (RLS), covering branch-, chain-, and account-level data isolation.
- Ran a full security and performance audit (RLS gap analysis, unfiltered realtime subscriptions, cron/backup scheduling, capacity modeling) and produced a hardening roadmap raising the platform's readiness score from ~6/10 to a projected 8+/10.
- Developed the tiered pricing model and financial plan (per-branch pricing, break-even and margin analysis, VAT-threshold planning) and a competitive analysis against 7 Egyptian pharmacy-software incumbents ahead of pilot sales.

LegalMind AI — Multi-Agent Legal Operating System — Python, Vector DB, LLM Orchestration

- Designed a multi-agent architecture with 11 specialist agents and hybrid (vector + keyword) search for an AI legal assistant targeting MENA, US, UK, and EU law firms.
- Produced a full build specification, design-principles document, and a UAE-market test pack; evaluated GLM-5.2 for generative agent tasks and selected an embedding strategy (BGE) and OCR approach for document-heavy workflows.

FitAI — Adaptive Fitness & Nutrition App — React, TypeScript, Node/Express, Firebase

- Built a mobile-first web app that generates personalized workout and meal plans using an LLM, combining survey data, uploaded body-composition PDFs, and live weather data.
- Designed a v2 adaptive quarterly planning system with calendar-based tracking, monthly AI-driven plan adjustments, and a safety interrupt for injuries or goal changes.

IoT Fleet Monitoring Platform — Multi-Tenant · Realtime · GPS Mapping

- Designed and led development of a multi-tenant monitoring platform for IoT field sensors (smart drains, smart bins) with live GPS mapping of deployed units.
- Architected role-based dashboards for 4 user types (Admin, Manager, Worker, Viewer) and automated stakeholder reporting; currently load-testing ahead of onboarding paying customers.

AI-Powered Google Workspace Assistant — n8n, Gmail API, Google Forms

- Built an automated assistant classifying incoming emails by intent (marketing, support, spam) and auto-generating replies, plus a Forms-to-stakeholder notification pipeline running end-to-end without manual intervention.

FEATURED DATA SCIENCE & MACHINE LEARNING PROJECTS

- Mask Detection (Computer Vision): Trained a TensorFlow CNN to detect face masks in images, ~91% accuracy.
- Earthquake Damage Prediction (Nepal): Logistic regression and decision-tree models on structural data from SQLite, identifying and correcting discriminatory bias in the dataset.
- Volatility Forecasting (India): GARCH model over API-sourced stock data, served through a custom API.
- Bankruptcy Prediction (Poland): Random forest and gradient boosting models with resampling to address severe class imbalance.
- Customer Segmentation (US): K-means clustering with PCA, delivered via an interactive Plotly Dash dashboard.
- Air Quality Forecasting (Nairobi) · Real Estate Regression (Buenos Aires) · Housing Price Analysis (Mexico, 21,000+ records): ARMA time-series modeling, regression pipelines with missing-value imputation, and price-driver analysis.

GRADUATION PROJECT

Smart Drain — IoT Structural Health Monitoring — 2024

- Designed an IoT system measuring street-drain water levels for early flood/damage detection, with multi-level sensors, color-coded alerting, and MQTT-based real-time transmission to a Firebase-backed dashboard, powered by solar + battery backup.

EDUCATION

Bachelor of Computer Science

Arab Academy for Science, Technology & Maritime Transport · 2020 – 2024 · GPA: 3.1 (Very Good)

Diploma in Machine Learning & Artificial Intelligence

Amit Learning · 2023 – 2024 · Grade: Excellent (95%)

CERTIFICATIONS

- Machine Learning & AI — Amit Learning (2024)
- AI Lab: Deep Learning for Computer Vision — WorldQuant University (07/2024 – 12/2024); CNNs, image classification, object detection, and facial recognition with TensorFlow and PyTorch.

TECHNICAL SKILLS

Core: Python, SQL, TypeScript/JavaScript, LLM API Integration (Claude, GLM), Multi-Agent Orchestration, PostgreSQL (Supabase, Row-Level Security), React, FastAPI

Familiar with: Java, PyTorch, TensorFlow, scikit-learn, MCP Tool Protocols, Vector Search (pgvector), RAG, Firebase/Firestore, MongoDB, SQLite, Celery, pandas, ETL Pipelines, Vite, JavaFX, Streamlit, Plotly Dash, Vercel, Render, MQTT/IoT, Twilio, n8n, Google Workspace API

LANGUAGES

Arabic — Native · English — Professional Working Proficiency